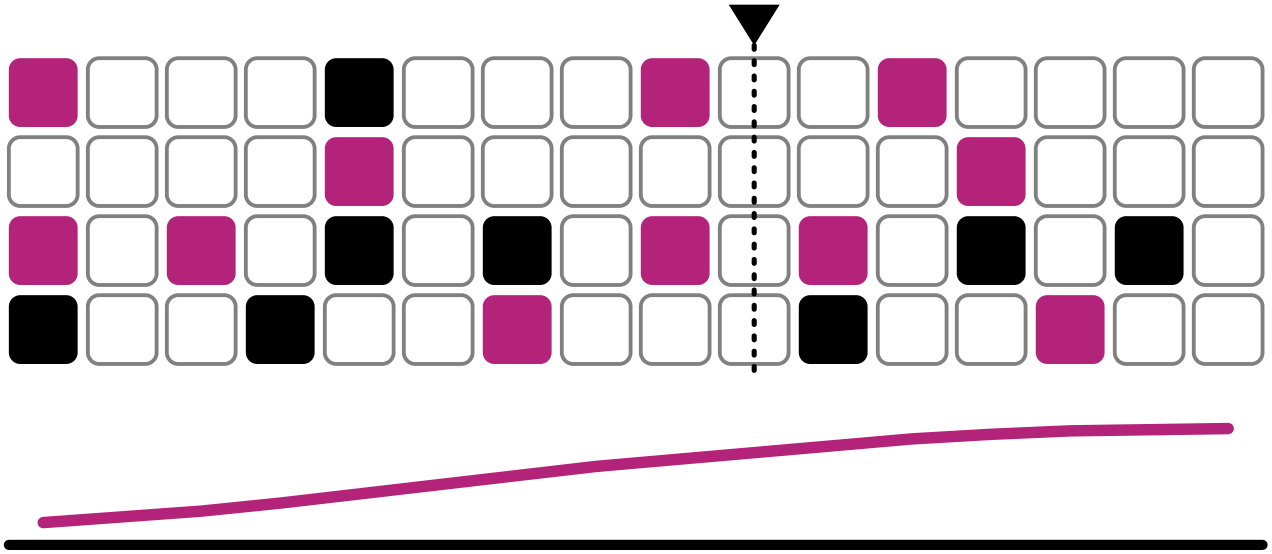
HUDSON BEAT



PROGRAM A GROOVE THAT EVOLVES
PATTERN PRODIGY

DMP2 · step sequencer · intro slides

GRADES 9–12 · 7 CLASS PERIODS · 24 PAGES + 20-SLIDE DECK

PATTERN PRODIGY

How to use this resource

START HERE

Teacher

WHAT THIS IS ·

A 7-class-period structured project for high school music technology. Each student builds a 30–60 second groove in Logic with the Step Sequencer and Pattern Regions: drums, a bass or synth and one more element, with probability, velocity and automation so the patterns evolve. They bounce it, reflect and present it with Padlet questions for feedback. It is the project from my Digital Music Production 2 course (type: structured; focus: Step Sequencer), and the projects in that course can be taught in any order.

QUICK START ·

1. Read the pacing guide (page 5) and my teacher notes (pages 6–7).
2. Set up the Presentation Padlet, practise the hi-hat demo and decide which features count (page 4).
3. Print the student pages (pages 13–23), one set per student.
4. Open the intro deck and read the speaker notes for Class 1.

THE INTRO DECK ·

Every slide has a class: Class 1 (slides 1–7), Class 2 (slides 8–10), Class 3 (slides 11–13), Class 4 (slides 14–15), Class 6 (slides 16–18) and Class 7 (slides 19–20). Class 5 is a working day with no new slide. Every slide has speaker notes.

Fonts: the deck uses Oswald and Source Sans 3 (free on Google Fonts, built into Google Slides). If PowerPoint or Keynote shows a different font, install the two fonts or present the PDF copy, which shows the intended look.

WHAT'S INSIDE ·

PART	PAGES	USE
Teacher pages	pages 2–12	How to use, overview, setup, pacing, teacher notes, what changed, Logic quick reference, pre-built pattern and Harder variation, worked example (2 pages)
Student pages	pages 13–23	Pattern grids, Feature tracker, Arrangement plan, Listen-back sheet, project sheet (3 pages), rubric, vocabulary, feedback planner, reflection
Intro deck (separate file)	deck	20 slides, .pptx + PDF, speaker notes on every slide

NO AUDIO INCLUDED

No audio, video, songs, artists or albums are named or included. Students make every sound in their tracks themselves, from the instruments and kits in Logic.

MAKE IT FIT YOUR CLASS ·

- **Easier or harder:** my variations on page 6: a pre-built drum pattern (a chart to enter is on page 10), or 90 seconds with pattern alternates and 2 types of step automation.
- **Feedback board:** my sheets use Padlet for the three questions; any class board works.
- **Any sounds:** the pages name no kit or instrument beyond Logic's own, so students fill in their own.
- **Grading:** my rubric, /25, the average of the student's and your score (page 20).

IN ONE SENTENCE

Students program a groove on a grid, then learn that probability, velocity and automation are how a loop evolves.

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PATTERN PRODIGY

Project overview

OVERVIEW

DMP2 · Structured

FROM MY DMP2 PROJECT MENU

Pattern Prodigy · Structured · Step Sequencer
Master Logic's Step Sequencer with automation, probability, and velocity to build evolving grooves.

WHAT STUDENTS MAKE ·

One original loop-based groove, 30–60 seconds long, built from Pattern Regions in the Step Sequencer: at least one drum kit, one bass or synth and one additional element. It uses at least three different Step Sequencer features, such as probability, velocity and step automation, and a mix with balance, clarity and space. They hand in a bounced .mp3 or .wav and the zipped .logicx project, plus a Padlet post with three feedback questions, then play the piece and present it.

WHAT THIS PROJECT REINFORCES ·

Introduces the Step Sequencer workflow. Builds rhythm programming, automation, and arrangement skills.

STANDARDS ·

Supports the National Core Arts Standards for Music (Music Technology, high school): generating and developing ideas (MU:Cr1.1, MU:Cr2.1), presenting work (MU:Pr6.1) and responding to it (MU:Re7.2).

GOALS ·

- Learn to create and manipulate Pattern Regions for drums, bass, and synths
- Use probability, velocity, and step automation to add organic variation
- Apply EQ, panning, and plug-ins for a balanced, polished sound
- Explore texture and form through pattern layering and evolving loops

AT A GLANCE ·

Grades	9–12: the second-year course, after a first year of Logic Pro basics
Length	7 class periods: explore and build drums, add melody, three working days, mix and bounce, present
Software	Logic Pro: Step Sequencer, Pattern Regions, automation, mixing, bounce
You supply	Macs with Logic Pro, headphones, a feedback board and a place to submit files
Assessment	My rubric, 5 categories scored 1–5, /25; final grade = average of the student's and the teacher's score (page 20)

BEFORE THEY START

Students should already know how to open a project, add and play software instruments, use the Mixer, and save, bounce and zip in Logic. The Step Sequencer is new: the quick reference (page 9) shows where its controls live.

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PATTERN PRODIGY

Setup and what to watch

SETUP

Teacher

YOU NEED ·

- **A Mac with Logic Pro** for each student, and **headphones**. Speakers help for the final listen.
- **A place to submit** the .mp3 or .wav and the zipped .logicx (your class page).
- **The Presentation Padlet**, set up before Class 1 (yours to set up; none is included).
- **A hi-hat demo** ready for Class 1 or 2 (my third struggle): a hi-hat row with a few steps at 25–50% probability, looped.
- **A rule for "at least 3 features"** (requirement 2): which features count. Write it on the board and on page 14.
- **Printed student pages**, one set each (pages 13–23).

BEFORE CLASS 1 ·

- [ ] Practise opening a Pattern Region and the Step Sequencer on a classroom Mac.
- [ ] Set up the Presentation Padlet in your class page.
- [ ] Decide which features count, and whether to offer the pre-built pattern (page 10).
- [ ] Check that a student computer opens Logic and plays back through headphones.

WHAT TO WATCH (MY STRUGGLES) ·

Students use the Step Sequencer only for drums, patterns sound static, probability is hard to see, and the short length makes students rush. Works well for students drawn to electronic, lo-fi, or beat-driven music.

KIT SUGGESTIONS, NOT PART OF MY SHEETS

- Features that count: probability, velocity, step automation, mute and unmute, and pattern alternates are the ones my notes and tips name. Add others if you like.
- Ask while students work: which feature changed what you just heard? Which step would you change next?
- Plan the bars first. Thirty seconds is only about 15 bars at 120 BPM, so a short piece goes fast.

SOUNDS THEY DID NOT MAKE ·

A KIT SUGGESTION

Tell students to build their grooves from the instruments and kits in Logic, and to hand in only their own track.

KEEP THE VOLUME MODERATE

Loud kicks and bass are easy to overdo in headphones. Tell students to start low and to check the final mix at a normal level.

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PATTERN PRODIGY

Pacing guide · 7 class periods

PACING

Teacher

7 class periods, as on my project menu. The “My plan” column is my day-by-day pacing guide as I wrote it; the other columns show what to put on the screen and in students’ hands.

CLASS	MY PLAN	TEACHER MOVES	DECK	STUDENT PAGES
CLASS 1	**Class 1:** Introduction to Step Sequencer and Pattern Regions. Guided exploration.	Slides 1–7: hook, what the project is, the steps, five words, steps 1 and 2. Hand out the project sheet and the Pattern grids. Guide the class through a first drum pattern; no audio is included.	Slides 1–7	p. 13, p. 17, p. 18, p. 19, p. 21
CLASS 2	**Class 2:** Build drum/percussion patterns. Begin adding melodic layers.	Slides 8–10: reading a pattern grid, not only drums, step 3. Demo a melodic pattern early (my first struggle). Check each student has a drum groove.	Slides 8–10	p. 14
CLASS 3	**Classes 3–5:** Working days — explore automation, probability, variation.	Slides 11–13: four ways to evolve, step 4 and the worked example on paper. Hand out the Arrangement plan. Then work time.	Slides 11–13	p. 15
CLASS 4		Slides 14–15: probability is your friend, and mix and refine. Hand out the Listen-back sheet. Then work time.	Slides 14–15	p. 16
CLASS 5		Full working day: automate, vary, arrange. Ask each student to loop one bar and say which step they would change next.	No new slides	—
CLASS 6	**Class 6:** Final mix, bounce. Prepare presentations.	Slides 16–18: bounce, what to turn in, the rubric. Students post their three Padlet questions.	Slides 16–18	p. 20, p. 22
CLASS 7	**Class 7:** Presentations with Padlet Q&A peer feedback.	Slides 19–20: presentation day and the close. Every student presents and answers 2 of each classmate’s questions. Reflection (page 23) follows the presentations.	Slides 19–20	p. 22, p. 23

THE ONE THING TO SHOW ON CLASS 1

Put a hi-hat on a grid and give a step a 50% chance (my third struggle). Loop it for a few bars so the class hears it change. Do the same on a bass row so it is not only for drums.

CLASSES 3–5

Working days; Class 5 has no new slide. Circulate with one question: which feature changed the sound you just heard?

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WHAT THIS PROJECT REINFORCES ·

Introduces the Step Sequencer workflow. Builds rhythm programming, automation, and arrangement skills.

COMMON STUDENT STRUGGLES ·

- Students only use the Step Sequencer for drums. Demo melodic patterns early.
- Patterns sound static. Push automation and probability.
- Students don't understand probability. Show a live example on a hi-hat.
- The short length makes students rush.

SUGGESTED VARIATIONS ·

Easier: Provide a pre-built drum pattern as a starting point.

More Advanced: Extend to 90 seconds. Require pattern alternates and 2 types of step automation.

WHERE THIS KIT HELPS WITH EACH STRUGGLE

- Only drums: slide 9 and the bass and synth rows on the Pattern grids (page 13).
- Static patterns: the Feature tracker (page 14) and slide 11 show four ways to make a pattern change.
- Probability: the hi-hat demo on page 4 and slide 14, with the dashed steps on the worked example (page 11).
- Rushing: the Arrangement plan (page 15) has a length helper, so students plan their bars first.

WHERE THIS KIT HELPS WITH THE VARIATIONS (KIT SUGGESTION)

- Easier: a 16-step chart of a drum pattern to enter, standing in for the pre-built pattern (page 10).
- More advanced: the worked example has an alternate pattern and step automation (page 12); page 10 lays out 90 seconds.

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SUGGESTED RESOURCES ·

MY LIST	IN THIS KIT
Logic Step Sequencer Getting Started (YouTube)	Kept as I wrote it. No video or link is included. The Logic quick reference (page 9) says where each Step Sequencer control lives, and the Pattern grids (page 13) give students a place to plan.
Live Pattern Recording tutorial (YouTube)	Kept as I wrote it. No video is included. Recording a pattern live is one way to fill a grid; the quick reference (page 9) mentions it.
Step Sequencer for Automation tutorial (YouTube)	Kept as I wrote it. No video is included. The Feature tracker (page 14) shows what to try and the example (page 12) shows the result.

HOW TO GRADE ·

- My rubric: five categories, each scored 1–5, /25 (page 20). Students score themselves first; the final grade is the average of their score and yours.

Suggestions from this kit, not part of my rubric:

- Creativity & Originality: ask what groove the student was after, then listen for what makes it theirs (my tip: start with your drums, then build).
- Technical Skill: open the project and count the Step Sequencer features against the Feature tracker (page 14). Is each one audible?
- Composition & Structure: compare the piece with the Arrangement plan (page 15). Does it build and evolve, or only stack?
- Sound Design & Mix Quality: listen for balance and space (requirement 4). Can you hear each pattern?
- Documentation & Presentation: the Feature tracker, the Arrangement plan, the Reflection page (page 23) and the Padlet questions are the evidence.

REFLECTION · ADDED BY THIS KIT ·

The rubric mentions reflection, but the project sheet has no reflection task, so this kit adds one (page 23): a written or oral conversation about how you did on the project and what you would do differently next time. Students write it or talk it through with you; either way they answer the four prompts. Allow about 10 minutes after the presentations. It adds no points and no rubric row; use it as evidence when you score Documentation & Presentation.

NOTES / OBSERVATIONS ·

Works well for students drawn to electronic, lo-fi, or beat-driven music.

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PATTERN PRODIGY

What changed from the original

CHANGES

Teacher

This kit is built from three of my documents: the **Project Sheet & Rubric**, the **Teacher Notes** and the **Pattern Prodigy** entry on my DMP2 project menu. This page lists every change to my wording, so you know exactly what is mine and what was added.

WORDING CHANGES ·

- My LMS name is written as “your class page” (checklist items 1 and 2).
- Page numbers are added to steps 1, 4 and 5 (the Pattern grids, Feature tracker, Arrangement plan and Listen-back sheet).
- My Easier variation (a pre-built drum pattern) is a Logic project, which is not included. It is laid out as a 16-step chart on paper (page 10) so you can enter it in a few minutes.
- My Harder variation (90 seconds, pattern alternates, 2 types of step automation) is laid out on page 10. The sheet does not say which features count toward “at least 3”; the Feature tracker lists the ones my notes and tips name (page 14).
- My resources are three YouTube tutorials. They are listed as I wrote them; no tutorial, video or link is included.
- My pacing guide’s “Classes 3–5” row spans the Class 3, 4 and 5 rows, so each class can name its slides.
- Layout only: the rubric’s Name and Date lines moved to the top; its Notes box became ruled lines.
- The intro deck shortens my wording (full wording on pages 17–19): slide 3 shortens my overview and goals, slide 6 shortens my five vocabulary definitions, slide 7 shortens steps 1 and 2, slide 9 shortens my first struggle and tip 1, slide 10 shortens step 3 and requirement 3, slide 12 shortens step 4 and requirement 2, slide 14 shortens tips 2 and 3 and my second and third struggles, slide 15 shortens step 5, tip 4 and the Harder variation, slide 16 shortens steps 6 and 7, slide 17 shortens the submission checklist, and slide 19 shortens step 7.
- **Reflection added (page 23).** My rubric mentions reflection, but the project sheet has no reflection task. This kit adds one, following my definition: a written or oral conversation about how the student did on the project and what they would do differently next time. It adds no points and no rubric row.

KEPT AS I WROTE IT ·

Everything else from my three documents is kept as I wrote it: the overview, goals, five vocabulary terms, four requirements, steps (apart from the page numbers), tips, checklist (apart from the class page) and file name (“Your Name – Pattern Prodigy”), the /25 rubric, what the project reinforces, my four struggles, the variations, my closing note and the project menu entry on page 3. No rubric row is added and the requirements are unchanged.

NEW IN THIS KIT ·

Everything else is new and supports my sheets without changing them: the how-to, overview and setup pages, the teacher moves in the pacing guide, the grading suggestions, the suggested list of features that count (a kit suggestion on page 4), the Logic quick reference, the pre-built pattern chart and Harder variation page, the worked example (an invented 50-second piece, its patterns, sections, layers and lanes, on paper, not audio), the Pattern grids, Feature tracker, Arrangement plan and Listen-back sheet, the vocabulary and presentation pages and the intro deck. None adds a requirement or a rubric row. The one addition that asks something of students is the Reflection page (page 23), explained under Wording changes; the rubric is unchanged.

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PATTERN PRODIGY

Logic quick reference

SETUP

Teacher + student

Where to find the tools a Pattern Prodigy project is likely to use. Demonstrate what your class needs on Classes 1 and 2 and then let students work. Names are from current Logic Pro versions; labels move between versions, so check yours.

New project and tracks	File › New…, then add tracks with Track › New Tracks…: a Software Instrument track for drum kits, bass and synths.
Drum kit and sounds	Choose a drum kit from the Library (Y) on a Software Instrument track. Kits that show a grid of pads, such as Drum Machine Designer, are the easiest to read in a sequencer.
Pattern Region and Step Sequencer	Create a Pattern Region on the track (look for the Pattern Region option when you make or draw a region) and double-click it to open the Step Sequencer in the editor area. Click a step to turn it on; click again to turn it off.
Bass and synth rows	A melodic track can use the Step Sequencer too: each row is a pitch, so the steps you click make a bass line or a synth arp. This answers my first struggle.
Velocity and probability	Use the controls along the bottom or side of the Step Sequencer to choose which value you edit for each step. Velocity is how hard a step hits; probability is its chance of playing. Try it on a hi-hat.
Step automation	Pick a parameter, such as a filter cutoff, and set a value for each step. The values repeat every time the pattern loops.
Mute and unmute	Each row has a mute control. Muting and unmuting rows as the piece plays is the simplest way to change the arrangement (my fourth tip).
Alternates	For my Harder variation, put a second Pattern Region with a different pattern after the first. Check whether your version also offers pattern sets inside one region.
Live recording	You can also play a pattern in live and let Logic fill the steps. Quantize or edit the steps afterwards.
Automation, EQ, pan, plug-ins	Press A to show track automation. Add Channel EQ for balance, use each track's Pan knob, and add plug-ins such as a reverb or filter. Press X for the Mixer.
Length check and bounce	Bars × 240 ÷ tempo = seconds. File › Bounce › Project or Section (Command-B), MP3 or WAV. In Finder, Control-click the .logicx and choose Compress.

VERSION NOTE

Menu names are from current Logic Pro versions; older versions may differ slightly. Logic Pro is a trademark of Apple Inc., named only to describe what the project needs. This resource is not affiliated with Apple.

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PATTERN PRODIGY

Pre-built pattern and my Harder variation

TEACHER

Easier + Harder

My Easier variation: provide a pre-built drum pattern as a starting point. A Logic project is not included in this kit, so here is the pattern as a chart. Enter it in a Pattern Region in a few minutes, or have students enter it, with a kit you have the right to use.

PRE-BUILT DRUM PATTERN: ONE BAR, 16 STEPS ·

Step	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Kick	■								■							
Snare					■								■			
Closed hi-hat	■		■		■		■		■		■		■		■	
Percussion				■								■				

1. New Software Instrument track with a drum kit; create a Pattern Region and open the Step Sequencer.
2. Turn on the steps shown in the chart. Kick on 1 and 9, snare on 5 and 13, the hi-hat on every odd step, and percussion on 4 and 12.
3. Loop it for a few bars and save it as “Starter”. Students add bass, synth and variation on top.

MY HARDER VARIATION, LAID OUT ·

MY OPTION	HOW TO USE IT
Extend to 90 seconds	That is 36 bars at 96 BPM or 45 bars at 120 BPM (page 15 has the length helper). The worked example (page 11) is 50 seconds, so each section is about twice as long.
Pattern alternates	Students make a second pattern (an A and a B) and switch between them, as the worked example does in its Build.
2 types of step automation	Two different parameters set per step, for example a filter cutoff on a synth and a velocity or gate pattern on a drum row. The worked example uses one (page 12).

HOW TO USE THIS PAGE

Offer the chart to students who freeze at the empty grid; nobody has to take it. Offer the Harder variation to students who finish early. Both are your choice, as in my notes. The 30–60 second length is the rule for everyone else (requirement 1).

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PATTERN PRODIGY

Worked example · an invented groove

EXAMPLE

Teacher

An **invented piece, "Night Shift", described on paper**: no audio is included. Students' projects will differ. 20 bars at 96 BPM, four sections, five layers, five Step Sequencer features (page 1 of 2).

1 · A WEAK PLAN

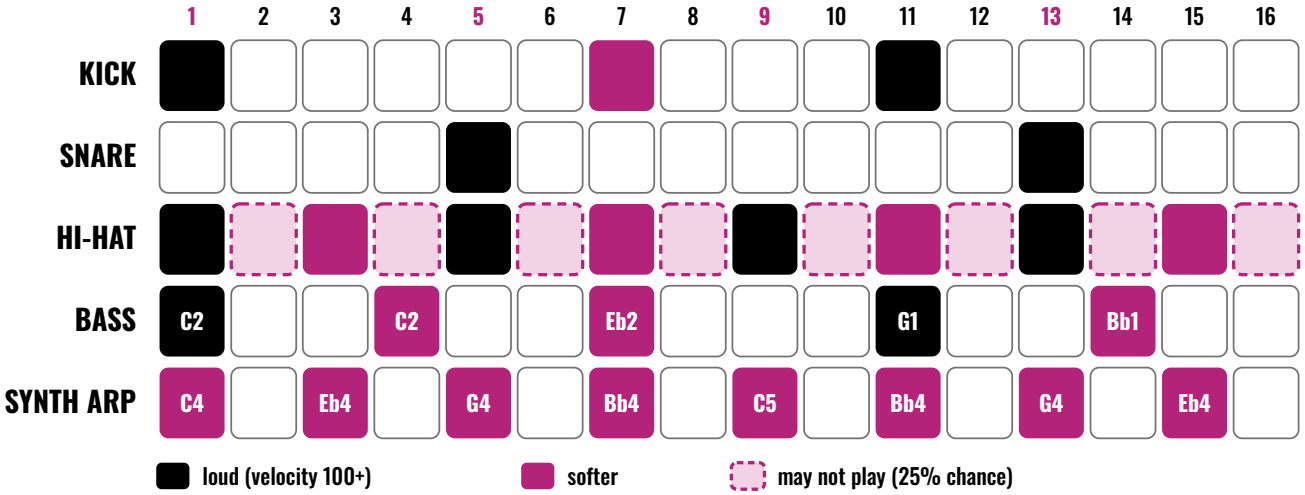
I will make a beat with drums and a bass.

It names no features and nothing says what will change.

2 · A CLEAR PLAN

I am making "Night Shift", a 50-second groove at 96 BPM. It opens with kick and snare, and the hi-hat row is unmuted at bar 3. A bass pattern arrives at bar 5 and a synth arp at bar 9. I will use five Step Sequencer features: probability on the hat's even steps, velocity accents, a cutoff on every step of the arp, mute and unmute, and an alternate drum pattern for the Build. Four automation lanes keep it moving, and in the last four bars it thins out.

3 · ONE BAR OF THE GROOVE: 16 STEPS ·



The bass plays C, Eb, G and Bb (C minor 7) on steps 1, 4, 7, 11 and 14, and the synth arp climbs C, Eb, G, Bb, C and comes back down on the odd steps. Every pitch is a chord tone, so the layers agree. Hat velocity: 100 on steps 1, 5, 9, 13 and 70 on the other odd steps; the even steps are ghosts at 25% probability.

4 · THE FIVE FEATURES IT USES ·

Probability	the hi-hat's even steps play 25% of the time; raised by hand to 60% in the Build
Velocity	hat accents on steps 1, 5, 9 and 13 (100) over softer ones (70); kick and snare at about 90 to 110
Step automation	the synth arp's filter cutoff has its own value on every step, rising from 15 to 97
Mute / unmute	the hat row is unmuted at bar 3; the snare row is muted at bar 19
Pattern alternate	a second drum pattern, B, plays in bars 13–16 and then the first returns

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PATTERN PRODIGY

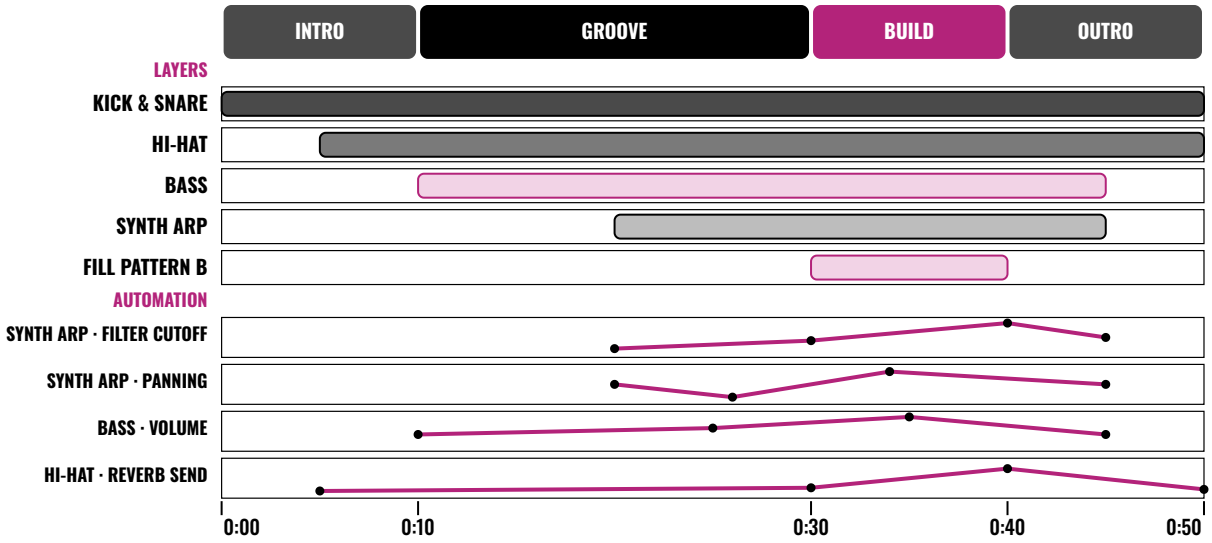
EXAMPLE

Worked example · the map

Teacher

"Night Shift", page 2 of 2: the map of the whole piece, then the numbers.

5 · THE MAP: 0:50, FOUR SECTIONS, FIVE LAYERS, FOUR LANES ·



6 · SECTIONS

Intro	bars 1–4	kick and snare; the hat row is unmuted at bar 3
Groove	bars 5–12	the bass joins at bar 5 and the arp at bar 9; the arp's cutoff begins to open
Build	bars 13–16	drum pattern B plays; hat ghost steps rise to 60%; the cutoff opens fully
Outro	bars 17–20	pattern A returns; the bass and arp leave at bar 19, the snare row is muted, the reverb falls

7 · LAYERS (IN TO OUT)

Kick & snare	a drum kit pattern region, 16 steps, looped	0:00–0:50
Hi-hat	the hat row, unmuted at bar 3, with ghost steps on probability	0:05–0:50
Bass	a bass pattern region from bar 5 to bar 18	0:10–0:45
Synth arp	a synth pattern region from bar 9 to bar 18	0:20–0:45
Fill pattern B	a second drum pattern for the Build, bars 13–16	0:30–0:40

8 · AUTOMATION LANES

Synth arp · filter cutoff	0:20–0:45
Synth arp · panning	0:20–0:45
Bass · volume	0:10–0:45
Hi-hat · reverb send	0:05–0:50

9 · CHECKING THE NUMBERS

One bar of 4/4 at 96 BPM lasts 240 ÷ 96 = 2.5 seconds, so 20 bars = 50 seconds, inside my 30–60 second range. The sections are 4 + 8 + 4 + 4 bars = 20. Five features, more than my minimum of three. **Three Padlet questions (invented):** 1. Which pattern did you hear change first? 2. Where did the hi-hat sound most alive, and why? 3. What would you take out to make the Build hit harder?

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PATTERN PRODIGY

Pattern grids

STUDENT

Classes 1–2

Name _____ Date _____

Steps 1 to 3: open the Step Sequencer and experiment with basic drum patterns, create your core drum and percussion patterns, then program bass and synth patterns that complement your groove. Colour or mark the steps you turn on. The Step Sequencer is a tool that lets you program rhythms and melodies by toggling steps on a grid. Each grid is one bar of 16 steps; beats start at 1, 5, 9 and 13.

PATTERN A · DRUMS AND PERCUSSION

Step	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Kick																
Snare																
Hi-hat																

Steps with lower velocity or probability: _____

PATTERN B · BASS (WRITE THE NOTE UNDER EACH STEP)

Step	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Bass note																
Note name																

PATTERN C · SYNTH OR ANOTHER ELEMENT

Step	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Synth note																
Note name																

MY GROOVE

Tempo (BPM): _____ **The feel I want:** _____

The row I will use for something other than drums: _____

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PATTERN PRODIGY

Feature tracker

STUDENT

Classes 2–5

Name _____ Date _____

Step 4: use probability, velocity, and automation to make patterns evolve. Requirement 2: use at least 3 different Step Sequencer features. Probability is a feature that gives each step a chance of playing or not, adding randomness. Velocity is how hard a note is played. Fill in the row each time you use a feature.

The features my teacher says count: _____

FEATURE	WHERE I USED IT (TRACK, PATTERN, BARS)	THE SETTING I CHOSE	WHAT I HEAR CHANGE	DONE?
Probability				
Velocity				
Step automation				
Mute / unmute				
Pattern alternate				
Another feature:				

CHECK

- ☐ I used at least 3 different Step Sequencer features (requirement 2).
- ☐ I can point to each one and play the moment it changes.
- ☐ I used the Step Sequencer for more than drums (a bass or synth row).

The feature that changed my groove most: _____

A feature I want to try next: _____

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PATTERN PRODIGY

Arrangement plan

STUDENT

Classes 3–5

Name _____ Date _____

Step 5: arrange into a cohesive piece. Use mute/unmute in the Step Sequencer to create arrangement changes. Shade the bars where each pattern plays, and write the layer's name at the left. Don't just stack patterns: think about how your piece builds and evolves.

Layer	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32

MY SECTIONS

Section 1: _____ **bars** _____

Section 2: _____ **bars** _____

Section 3: _____ **bars** _____

Section 4: _____ **bars** _____

LENGTH HELPER (BARS NEEDED)

BPM	30 S	60 S
80	10	20
96	12	24
120	15	30
128	16	32
144	18	36

My tempo: _____ **Bars:** _____ **Length: bars × 240 ÷ tempo =** _____ **seconds**

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PATTERN PRODIGY

Listen-back sheet

STUDENT

Classes 4–6

Name _____ Date _____

Step 5: apply EQ, panning, and plug-ins. Arrange into a cohesive piece. Requirement 4: demonstrate balance, clarity, and creative use of space in your mix. Loop one section and listen for each pattern.

LISTEN-BACK

The pattern I hear clearly: _____

The pattern that gets buried: _____ **What fixes it (EQ, pan, level):** _____

A bar that sounds exactly like the one before it: _____

STATIC OR CROWDED SPOTS ·

BAR OR TIME	WHAT FEELS STATIC, CROWDED OR MUDDY	WHAT I CHANGED

CHECKS BEFORE I BOUNCE

- ☐ I have a drum kit, a bass or synth, and one more element.
- ☐ At least 3 different Step Sequencer features are used and I can hear them.
- ☐ The piece builds and evolves; it does not only stack patterns.
- ☐ Mute and unmute change the arrangement at least once.
- ☐ The mix is balanced and clear, with room between sounds.
- ☐ The piece is between 30 and 60 seconds.

The one change that improved my groove most: _____

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Name _____ Date _____

PATTERN PRODIGY

Digital Music Production | *Structured Project* | 7 class periods

OVERVIEW ·

Become a pattern master. Use Logic's Step Sequencer and Pattern Regions to craft an original groove-driven composition. Create a 30–60 second piece that evolves over time through layering, automation, and creative use of sequencer features like probability and velocity.

GOALS ·

- Learn to create and manipulate Pattern Regions for drums, bass, and synths
- Use probability, velocity, and step automation to add organic variation
- Apply EQ, panning, and plug-ins for a balanced, polished sound
- Explore texture and form through pattern layering and evolving loops

KEY VOCABULARY ·

Step Sequencer	A Logic tool that lets you program rhythms and melodies by toggling steps on a grid
Pattern Region	A region in Logic that contains step sequencer data
Probability	A feature that gives each step a chance of playing or not, adding randomness
Velocity	How hard a note is played — higher velocity = louder and often brighter
Automation	Changing a parameter (volume, filter, pan, etc.) over time to create movement

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PATTERN PRODIGY

STUDENT

Project sheet · requirements and steps

Name _____ Date _____

REQUIREMENTS ·

- ☐ Length: 30–60 seconds
- ☐ Use at least 3 different Step Sequencer features
- ☐ Include at least one drum kit, one bass or synth, and one additional element
- ☐ Demonstrate balance, clarity, and creative use of space in your mix

STEPS / PROCESS ·

1. **Explore** — Open the Step Sequencer and experiment with basic drum patterns (Pattern grids: page 13)
2. **Build rhythms** — Create your core drum and percussion patterns
3. **Add melodic layers** — Program bass and synth patterns that complement your groove
4. **Automate and vary** — Use probability, velocity, and automation to make patterns evolve (Feature tracker: page 14)
5. **Mix and refine** — Apply EQ, panning, and plug-ins. Arrange into a cohesive piece. (Arrangement plan: page 15; Listen-back sheet: page 16)
6. **Finalize** — Bounce your track and prepare your presentation
7. **Present & Get Feedback** — Create your own Padlet post with 3 questions you want classmates to answer about your project. Present to the class. During classmates' presentations, answer at least 2 of their 3 questions on their Padlet post.

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PATTERN PRODIGY

Project sheet · tips and checklist

STUDENT

Name _____ Date _____

TIPS FOR SUCCESS ·

- Start with your drums. Once you have a groove that feels good, everything else falls into place.
- Probability is your friend — even a small percentage of randomness makes patterns feel more human.
- Don't just stack patterns. Think about how your piece builds and evolves.
- Use mute/unmute in the Step Sequencer to create arrangement changes.

SUBMISSION CHECKLIST ·

- ☐ Bounced .mp3 or .wav uploaded to your class page
- ☐ Zipped .logicx project file uploaded to your class page
- ☐ Files named correctly: "Your Name – Pattern Prodigy"
- ☐ At least 3 Step Sequencer features used
- ☐ Padlet post created with your 3 feedback questions
- ☐ Answered at least 2 questions on classmates' Padlet posts during presentations

NOTES ·

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PATTERN PRODIGY

Rubric

RUBRIC

/ 25

Name _____ Date _____

GRADING RUBRIC /25

Pattern Prodigy | Structured Project

Each category is scored 1–5. Your final grade is the average of your self-score and your teacher's score.

Category	1	2	3	4	5
CREATIVITY & ORIGINALITY	Minimal creative effort; relies on defaults or clichés	Some creative ideas but largely predictable	Shows originality; experiments with sound or style	Highly creative; unique approach with clear artistic intent	Exceptional creativity; innovative, bold, and expressive
TECHNICAL SKILL / DAW USE	Limited understanding of Logic Pro tools	Basic tool use with frequent errors	Competent workflow; uses tools correctly	Advanced use of Logic features and shortcuts	Expert-level command of multiple Logic systems
COMPOSITION & STRUCTURE	Disorganized; no clear form or direction	Some structure present but underdeveloped	Clear, cohesive form with logical sections	Engaging structure with dynamic progression	Professional pacing, flow, and musical arc
SOUND DESIGN & MIX QUALITY	Poor sound choices; unbalanced or muddy mix	Some variety but inconsistent levels and EQ	Good sound selection; mostly balanced mix	Strong sound palette; clean, well-balanced mix	Exceptional sound design; professional-quality mix
DOCUMENTATION & PRESENTATION	Missing or incomplete; no reflection	Minimal documentation with little insight	Satisfactory; provides basic detail and reflection	Well-documented; clear and thoughtful reflection	Exceptional; thorough insight and polished delivery

Student Score: _____ / 25	Teacher Score: _____ / 25
Final Grade = Average of Student + Teacher = _____ / 25	

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PATTERN PRODIGY

Key vocabulary

STUDENT

Vocabulary

Name _____ Date _____

The five words from your project sheet, with my definitions. Rewrite each in your own words, then show where it happens in your own project.

STEP SEQUENCER — A Logic tool that lets you program rhythms and melodies by toggling steps on a grid

In my own words: _____

Where it shows up in my track: _____

PATTERN REGION — A region in Logic that contains step sequencer data

In my own words: _____

Where it shows up in my track: _____

PROBABILITY — A feature that gives each step a chance of playing or not, adding randomness

In my own words: _____

Where it shows up in my track: _____

VELOCITY — How hard a note is played — higher velocity = louder and often brighter

In my own words: _____

Where it shows up in my track: _____

AUTOMATION — Changing a parameter (volume, filter, pan, etc.) over time to create movement

In my own words: _____

Where it shows up in my track: _____

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PATTERN PRODIGY

Present and get feedback

STUDENT

Classes 6–7

Name _____ Date _____

FROM YOUR PROJECT SHEET, STEP 7

Present & Get Feedback — Create your own Padlet post with 3 questions you want classmates to answer about your project. Present to the class. During classmates' presentations, answer at least 2 of their 3 questions on their Padlet post.

MY THREE FEEDBACK QUESTIONS ·

Ask about things you really want to know, not yes-or-no questions.

Question 1: _____

Question 2: _____

Question 3: _____

CLASSMATES' QUESTIONS I ANSWERED (AT LEAST 2) ·

CLASSMATE	THEIR QUESTION	MY ANSWER

BEFORE I PRESENT ·

- ☐ Files named "Your Name – Pattern Prodigy"; the .mp3 or .wav and zipped .logicx submitted
- ☐ Padlet post created with my 3 questions
- ☐ Rubric self-score done (page 20)

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PATTERN PRODIGY

Reflection · how did it go?

STUDENT

After presenting

Name _____ Date _____

WHAT A REFLECTION IS

A reflection is a written or oral conversation about how you did on the project and what you would do differently next time. Answer all four prompts in your own words. You choose how.

☐ **Written.** I will write my answers on this page.

☐ **Oral.** I will talk my answers through with my teacher. My teacher fills in the box at the bottom.

1 **What went well?** Name something in your project you are proud of, and say why it worked.

2 **What did you struggle with?** Say what was hard and what you tried to fix it.

3 **What would you do differently next time?** Be specific: a step, a choice, a sound or a tool.

4 **Look at the rubric (page 20).** Which category are you proudest of, and which would you work on first next time?

ORAL REFLECTION · TEACHER

Date: _____ **Prompts covered (1-4):** _____

Key points I heard: _____

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QUESTIONS?

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CREDITS

- Project sheet, rubric, teacher notes and intro deck: original work by Brian Lotze, Hudson Beat.
- Worked example (an invented 50-second piece with its patterns, sections, layers and automation lanes, described on paper), the grid, step, feature and arrangement drawings and layouts: original work, Hudson Beat.
- No audio, video, lyrics, album art, screenshots, presets, templates, links or chart data are included. No student work is included. Padlet is named only as the owner's own materials name it.
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